Threat Assessment Report

Executive summary for model robustness under adversarial attack

Model: microsoft/resnet-50

Attack Type: PGD
Images Evaluated: 10

Attack Success Rate	Original Confidence	Adversarial Confidence	Execution Time
100.0%	62.26%	99.90%	101.07s
			Accuracy drop: -37.64%

Threat Level Assessment

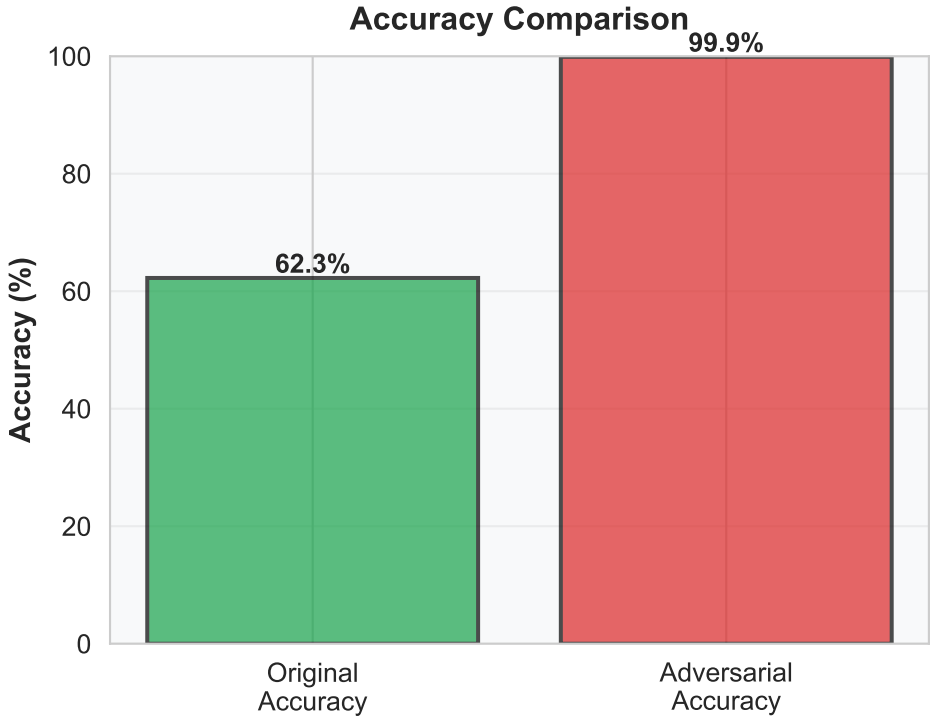
HIGH RISK

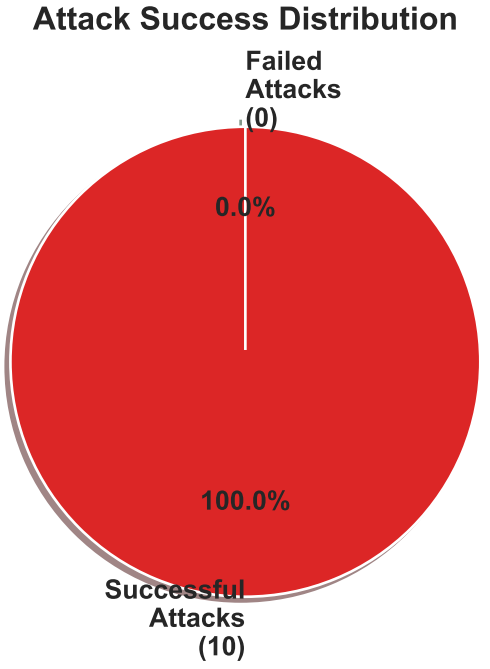
The PGD attack altered model predictions on 100.0% of the evaluated images. Average confidence shifted from 62.26% to 99.90%, indicating the current robustness posture is high risk.

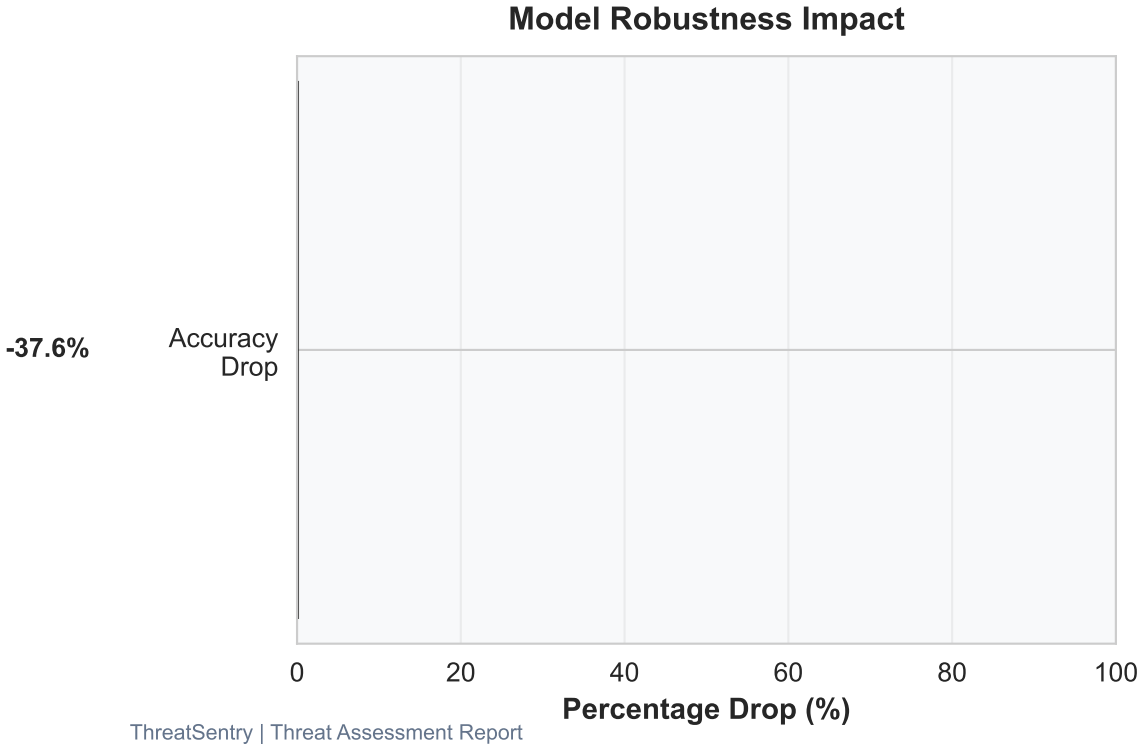
Assessment Narrative

Successfully executed PGD attack on model microsoft/resnet-50 using 10 test images. Attack success rate: 100.0%. Average original accuracy: 62.26%, Average adversarial accuracy: 99.90%. The attack successfully fooled the model in 10 out of 10 cases.

Detailed Analysis







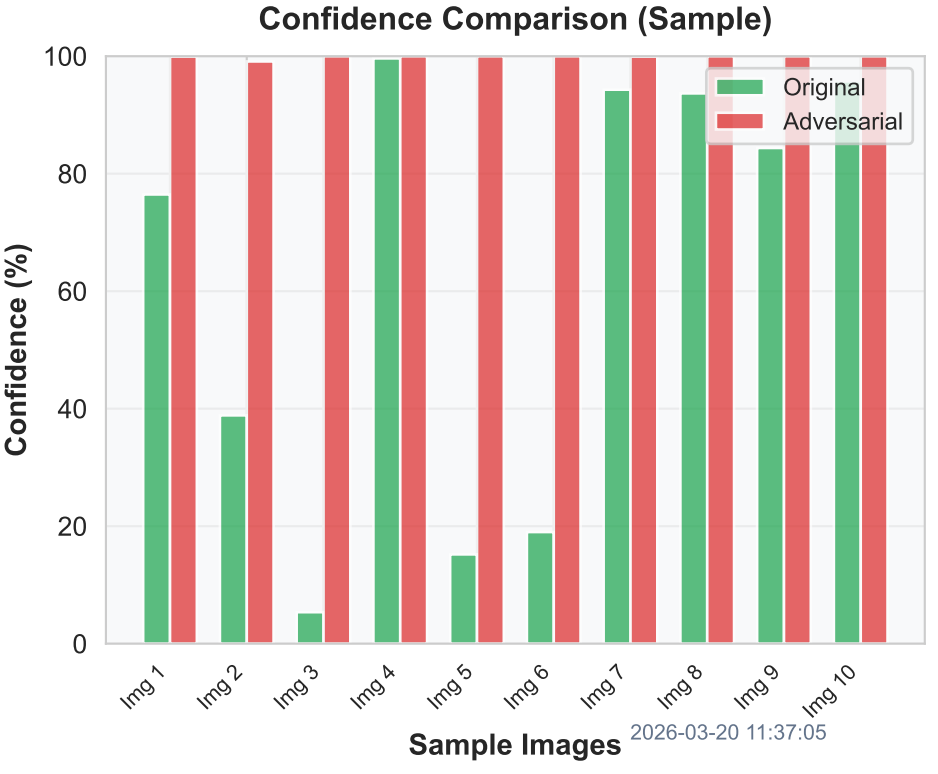


Image Evidence and Predictions

Images 1-4 of 10

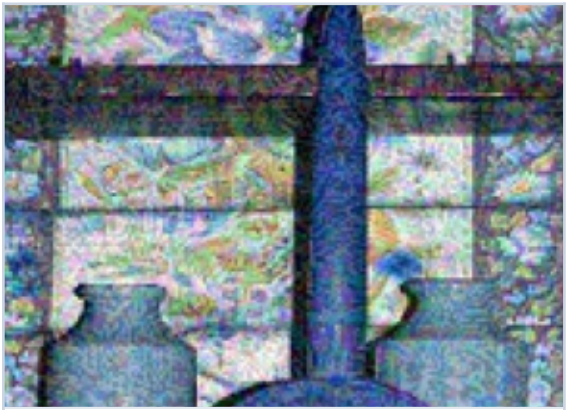
Attack Successful
Original image: reel.JPEG
Adversarial image: reel.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: reel
Original confidence: 99.62%
After attack predicted as: gasmask, respirator, gas helmet
Adversarial confidence: 100.00%

Attack Successful
Original image: ski.JPEG
Adversarial image: ski.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: ski
Original confidence: 94.30%
After attack predicted as: agama
Adversarial confidence: 99.93%

Attack Successful
Original image: butternut_squash.JPEG
Adversarial image: butternut_squash.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: brain coral
Original confidence: 76.47%
After attack predicted as: sea anemone, anemone
Adversarial confidence: 99.94%

Attack Successful
Original image: feather_boa.JPEG
Adversarial image: feather_boa.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: feather boa, boa
Original confidence: 15.20%
After attack predicted as: teddy, teddy bear
Adversarial confidence: 100.00%

Image Evidence and Predictions



Attack Successful
Original image: wooden_spoon.JPEG
Adversarial image: wooden_spoon.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: milk can
Original confidence: 84.38%
After attack predicted as: dumbbell
Adversarial confidence: 100.00%

Attack Successful
Original image: agaric.JPEG
Adversarial image: agaric.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: mushroom
Original confidence: 93.67%
After attack predicted as: electric ray, crampfish, numbfish, torpedo
Adversarial confidence: 100.00%

Attack Successful
Original image: space_shuttle.JPEG
Adversarial image: space_shuttle.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: space shuttle
Original confidence: 95.72%
After attack predicted as: thresher, thrasher, threshing machine
Adversarial confidence: 100.00%

Attack Successful
Original image: bakery.JPEG
Adversarial image: bakery.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: dining table, board
Original confidence: 5.36%
After attack predicted as: bath towel
Adversarial confidence: 100.00%

Image Evidence and Predictions

Attack Successful
Original image: tiger_beetle.JPEG
Adversarial image: tiger_beetle.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: isopod
Original confidence: 19.00%
After attack predicted as: cockroach, roach
Adversarial confidence: 100.00%

Attack Successful
Original image: Indian_elephant.JPEG
Adversarial image: Indian_elephant.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: tusker
Original confidence: 38.86%
After attack predicted as: cougar, puma, catamount, mountain lion, painter, panther, Felis concolor
Adversarial confidence: 99.11%

Recommendations

Security actions based on observed threat exposure

Assessment Summary

Successfully executed PGD attack on model microsoft/resnet-50 using 10 test images. Attack success rate: 100.0%. Average original accuracy: 62.26%, Average adversarial accuracy: 99.90%. The attack successfully fooled the model in 10 out of 10 cases.

Security Recommendations

1. Implement Adversarial Training

- Retrain your model with adversarial examples to improve robustness
- Use techniques like FGSM, PGD during training phase

2. Add Input Validation & Preprocessing

- Implement input sanitization and anomaly detection
- Use defensive distillation or feature squeezing

3. Deploy Ensemble Methods

- Use multiple models with different architectures
- Implement voting mechanisms for predictions

4. Continuous Monitoring

- Set up real-time performance monitoring
- Detect and alert on unusual prediction patterns

5. Regular Security Audits

- Conduct periodic threat assessments
- Stay updated with latest attack techniques